

REVIEW

Open Access



Application of artificial intelligence in medical imaging for tumor diagnosis and treatment: a comprehensive approach

Junyan Huang^{1*}, Yizhen Xiang², Shengqi Gan³, Linrong Wu³, Jiangyu Yan³, Dong Ye³ and Junjun Zhang^{4*}

*Correspondence:

Junyan Huang
2804092069@qq.com
Junjun Zhang
604988147@qq.com

¹Department of Radiology, the Second People's Hospital of Lishui, Wenzhou Medical University, Lishui, Zhejiang, China

²Digital Health Center, Berlin Institute of Health at Charité, Universitätsmedizin Berlin, Berlin, Germany

³Department of Otorhinolaryngology-Head and Neck Surgery, The Affiliated Lihuili Hospital, Ningbo University, Ningbo, China

⁴Department of Trauma Surgery, Yinzhou No.2 Hospital, Ningbo, Zhejiang, China

Abstract

This narrative review provides a comprehensive and structured overview of recent advances in the application of artificial intelligence (AI) to medical imaging for tumor diagnosis and treatment. By synthesizing evidence from recent literature and clinical reports, we highlight the capabilities, limitations, and translational potential of AI techniques across key imaging modalities such as CT, MRI, and PET. Deep learning (DL) and radiomics have facilitated automated lesion detection, tumour segmentation, and prognostic assessments, improving early cancer detection across various malignancies, including breast, lung, and prostate cancers. AI-driven multi-modal imaging fusion integrates radiomics, genomics, and clinical data, refining precision oncology strategies. Additionally, AI-assisted radiotherapy planning and adaptive dose optimisation have enhanced therapeutic efficacy while minimising toxicity. However, challenges persist regarding data heterogeneity, model generalisability, regulatory constraints, and ethical concerns. The lack of standardised datasets and explainable AI (XAI) frameworks hinders clinical adoption. Future research should focus on improving AI interpretability, fostering multi-centre dataset interoperability, and integrating AI with molecular imaging and real-time clinical decision support. Addressing these challenges will ensure AI's seamless integration into clinical oncology, optimising cancer diagnosis, prognosis, and treatment outcomes.

Keywords Artificial intelligence, Deep learning, Oncologic imaging, Radiomics, Tumour segmentation, Multi-modal fusion, Precision oncology

1 Introduction

Medical imaging plays a pivotal role in tumor diagnosis and treatment. Modalities such as X-ray, CT, MRI, and PET are fundamental for cancer screening, staging, and therapy assessment. However, the complexity of imaging data and subjective interpretation present challenges, including high-dimensional data, inter-observer variability, misdiagnosis, and uneven distribution of medical resources. Recent advancements in artificial intelligence (AI) have revolutionized medical imaging. Deep learning and machine learning techniques enable AI to identify complex patterns within vast medical imaging datasets,



© The Author(s) 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

significantly enhancing diagnostic accuracy and efficiency. Studies indicate AI excel in early detection and risk assessment of malignancies such as breast, lung, and prostate cancers, while also demonstrating potential in radiotherapy planning, personalized treatment, and follow-up management [1].

Despite its promise, AI adoption in medical imaging faces obstacles, including data quality issues, lack of interpretability, regulatory constraints, and ethical concerns. Understanding AI's advantages and challenges in oncologic imaging is essential for advancing precision medicine.

1.1 Definition and evolution of AI

Artificial intelligence (AI), first conceptualized in the 1950s, aims to simulate human cognition and reasoning. Early AI systems relied on rule-based logic and statistical machine learning, but recent advances in deep learning—such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and Transformer models—have enhanced AI's ability to automatically learn patterns from data [2]. In medical imaging, AI has evolved from basic enhancement and segmentation to intelligent diagnostics and personalized treatment. AI-driven analysis now detects lesions and compares them with radiologists' assessments, improving diagnostic accuracy. For example, AI in breast cancer screening identifies microcalcifications with greater consistency than traditional methods [3].

As computational power and data availability increase, AI applications have expanded into radiomics, computational pathology, virtual reality (VR), and augmented reality (AR), further enhancing precision medicine [4]. Additionally, multi-modal AI integrating imaging, clinical, and genomic data is improving tumor detection and treatment planning.

2 Applications of AI in medical imaging

AI has become a key focus in medical imaging, revolutionizing radiology, oncology, and precision medicine. Advancements in deep learning enable AI to analyze complex imaging data, enhancing screening accuracy, diagnostic efficiency, and optimizing workflows to reduce healthcare professionals' workload [5]. AI is widely used in image recognition, lesion detection, disease classification, radiomics, and image-guided interventions [6]. Tadiboina et al. [7] noted that AI not only improves disease detection but also helps clinicians develop personalized treatment strategies, advancing precision medicine [7].

In oncology, AI combined with radiomics extracts imaging features to predict tumor subtypes and treatment responses, aiding early diagnosis and targeted therapy. Radiomics pipelines are increasingly integrated directly into PACS/RIS infrastructures—enabling automated feature extraction, image retrieval, and annotation workflows within clinical systems—thus enhancing data traceability and reducing manual workload [8, 9] (Fig. 1).

2.1 Deep learning in imaging analysis

Deep learning (DL) has achieved significant progress in medical imaging analysis, excelling in image classification, object detection, segmentation, feature extraction, and pattern recognition. CNNs, in particular, have become the dominant technology in medical

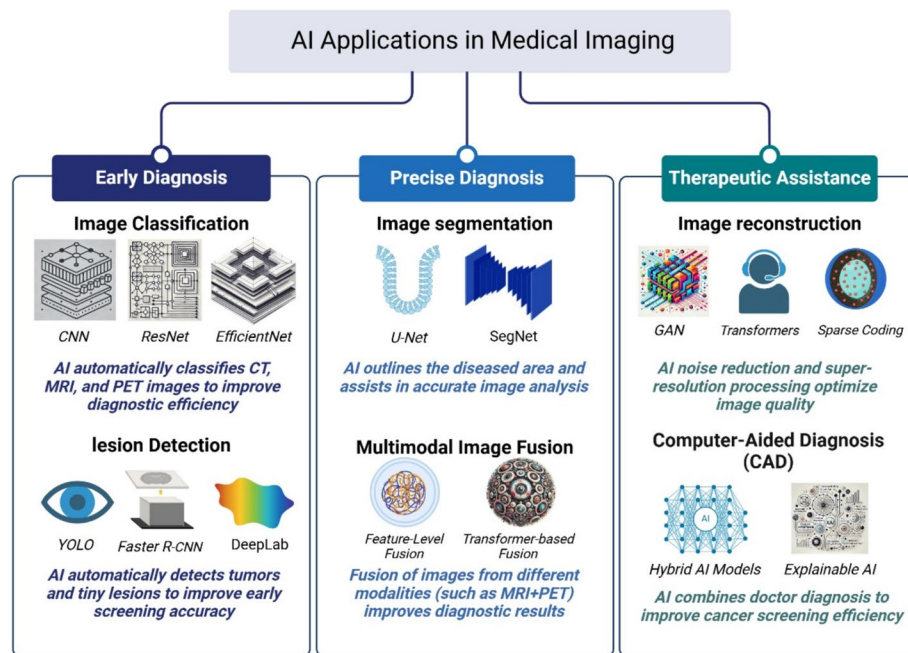


Fig. 1 AI Applications in Medical Imaging. This figure highlights the primary applications of AI in medical imaging, categorized into three key areas: early diagnosis, precise diagnosis, and therapeutic support. (1) Early Diagnosis: Encompasses image classification and lesion detection, enabling automated cancer screening and the identification of abnormalities in medical images (e.g., mammography, CT, MRI, PET). (2) Precise Diagnosis: Involves image segmentation and multimodal image fusion, improving tumor localization, boundary delineation, and diagnostic precision. (3) Therapeutic Support: Includes image reconstruction and computer-aided diagnosis (CAD), enhancing image quality and aiding clinical decision-making. Each application leverages specific deep learning methods (e.g., CNN, U-Net, GAN) and imaging data types

image processing, enabling the automated identification of complex imaging patterns and improving diagnostic precision [10].

2.1.1 AI in medical image classification

Litjens et al. [11] demonstrated that deep learning models trained on large, annotated datasets effectively classify medical images and detect pathological features [11]. They evaluated CNN performance on a dataset comprising over 10,000 mammograms from five Dutch hospitals, demonstrating that AI-powered breast cancer screening systems analyze mammographic images to identify microcalcifications, increasing early cancer detection rates. Additionally, deep learning enhances lung cancer detection in CT scans by improving the sensitivity and specificity of pulmonary nodule identification and risk assessment.

2.1.2 AI in medical image segmentation

Chen et al. [12] emphasized that AI-driven segmentation techniques are widely applied in tumor boundary recognition, organ segmentation, and lesion delineation [12]. Deep learning models such as U-Net and DeepLabV3+ have been successfully used in MRI and CT image segmentation, achieving high accuracy in brain tumor, lung lesion, liver cancer, and prostate cancer imaging. Similarly, Chen et al. [12] validated U-Net segmentation models on a cohort of 500 brain MRI cases across three tertiary-care centers, reporting an average Dice score of 0.89 but performance degradation in low-field MRI environments.

2.1.3 Transfer learning in medical imaging

Wang et al. [13] found that transfer learning significantly enhances deep learning model generalizability, especially in cases with limited labeled medical imaging datasets [13]. By leveraging CNNs pretrained on large-scale datasets such as ImageNet, transfer learning facilitates efficient adaptation to medical imaging tasks, improving performance in lung cancer screening, stroke detection, and retinal disease diagnosis.

2.1.4 AI in medical image reconstruction

Beyond classification and segmentation, AI has also advanced medical image reconstruction and enhancement [14]. AI-driven CT noise reduction and MRI super-resolution reconstruction improve image quality while minimizing scan duration and radiation exposure. For example, low-dose CT reconstruction algorithms based on deep learning reduce artifacts and enhance clarity, making them valuable for lung screening and orthopedic imaging.

2.1.5 Future perspectives

The application of deep learning in medical imaging continues to expand, moving toward multimodal integration and clinical decision support. As large-scale imaging datasets grow and computational capabilities advance, AI will play an increasingly pivotal role in medical diagnostics, disease prediction, treatment planning, and personalized medicine.

2.2 Multimodal imaging fusion

Multimodal Imaging Fusion (MMIF) combines data from MRI, CT, and PET to improve disease diagnosis and treatment. While single-modality imaging lacks comprehensive insights, fusing modalities enhances contrast, detail, and lesion detection [15].

Huang et al. [16] identified three fusion methods: pixel-level, feature-level, and decision-level fusion [16]. Pixel-level fusion uses mathematical transformations, like wavelet analysis, PCA, and deep learning, to merge grayscale information from different modalities. Feature-level fusion extracts key features from each modality and combines them with CNNs for better lesion recognition. Decision-level fusion, commonly used in CAD systems, integrates AI model predictions to boost diagnostic accuracy.

2.2.1 Applications of imaging fusion in tumor detection

Guo et al. [17] demonstrated the significant role of multimodal imaging fusion in tumor diagnosis and staging assessment [17]. In brain tumor diagnosis, PET imaging provides metabolic activity insights, while MRI offers precise anatomical details. Combining these modalities enhances tumor boundary detection. Similarly, in lung cancer screening, CT imaging provides high-resolution anatomical information, whereas PET imaging identifies metabolically active lesions, reducing false positives and improving early-stage cancer detection sensitivity.

2.2.2 Role of AI in multimodal imaging fusion

Saleh et al. [18] highlighted advances in AI-driven multimodal imaging fusion [18]. While traditional methods rely on manually designed features, AI-based approaches automatically learn latent features and integrate modalities more efficiently. For example, Generative Adversarial Networks (GANs) enable high-resolution PET-CT fusion,

enhancing lesion visibility, and Transformer networks improve contrast and boundary segmentation.

Khan et al. [19] forecasted that future fusion techniques will focus on greater automation and intelligence [19]. Key areas include AI-driven adaptive fusion algorithms, integrating multimodal imaging with biomarker data for personalized treatment, and real-time fusion for precise image-guided surgeries and radiotherapy. As computational power and large-scale datasets grow, multimodal fusion will become essential in precision medicine.

3 AI in tumor diagnosis and treatment via medical imaging

Recent AI breakthroughs have transformed medical imaging, enhancing tumor diagnosis and treatment. AI uses deep learning, radiomics, and computer vision to improve early screening, optimize treatments, and predict outcomes [20]. Unlike traditional methods that rely on radiologists' expertise, AI automates image analysis, detects subtle features, reduces misdiagnosis, and aids in precise clinical decisions [21].

AI in oncologic imaging covers image acquisition, lesion detection, and image-guided therapy (Fig. 2). Chen et al. [22] highlighted AI's role in optimizing tumor imaging parameters, reducing artifacts, and enabling multi-modal integration of CT, MRI, and PET scans [22]. AI-driven Computer-Aided Diagnosis (CAD) systems show high sensitivity in screening breast, lung, and prostate cancers. By extracting imaging features and integrating clinical and molecular data, AI aids in accurate staging, predicts treatment responses, and supports personalized treatment strategies.

As AI evolves, its integration with big data, genomics, and molecular imaging will refine tumor detection and personalize cancer management, leading to better patient outcomes.

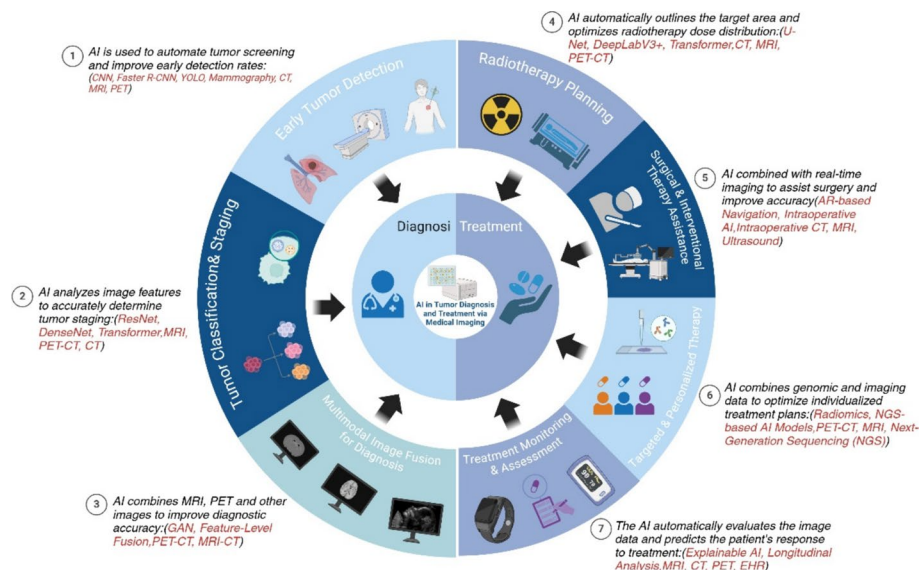


Fig. 2 AI applications in tumor diagnosis and treatment. This figure presents AI-driven applications in tumor diagnosis and tumor treatment via medical imaging. (1) Early Tumor detection; (2) Tumor classification & staging; (3) Multimodal image fusion for diagnosis; (4) Radiotherapy planning; (5) Surgical & Interventional therapy assistance; (6) Targeted & Personalized therapy; (7) Treatment monitoring & assessment

3.1 AI in tumor diagnosis through medical imaging

AI advancements in medical imaging have transformed tumor detection, diagnosis, and personalized treatment. Unlike traditional diagnosis, which relies on radiologists' assessments, AI offers objective, accurate, and efficient evaluations [23]. Deep learning (DL)-based models automatically extract imaging features, precisely identifying tumors and reducing human error, improving screening sensitivity and specificity [24].

AI has been applied in diagnosing cancers such as lung, breast, prostate, and brain tumors. Ijaz and Woźniak highlighted AI's role in optimizing CT, MRI, and PET imaging, and integrating radiomics and pathology data for tumor staging and molecular prediction [25]. AI-driven Computer-Aided Diagnosis (CAD) systems enhance diagnostic efficiency by automating image analysis and clinical decision-making [26].

As AI evolves, it increasingly supports multimodal imaging fusion, personalized treatment, and survival prediction. Future advancements will integrate AI with genomics and molecular imaging to improve cancer screening and treatment, advancing precision medicine.

3.1.1 AI-assisted early tumor detection

Early tumor detection is vital for improving cancer survival. AI automates medical imaging analysis, identifying early lesions and enabling timely intervention, enhancing screening accuracy and efficiency [27]. Deep learning models, especially CNNs, detect small tumors and anomalies in mammography, CT, and MRI scans with high precision, improving early diagnosis reliability. Kaur & Garg found AI effective in detecting early lung, breast, and liver cancers, particularly in CT-based lung screening, where it analyzes nodules and assesses malignancy risk [26]. AI also uses historical datasets to identify high-risk patients and assist with personalized interventions. Zheng et al. highlighted AI's role in breast cancer screening through MRI analysis, detecting complex patterns and early abnormalities [28]. AI systems improve detection rates and reduce false positives/negatives.

AI aids multimodal imaging fusion by integrating CT, MRI, and PET scans to extract critical features, refining early detection [25]. As AI technology and datasets expand, its role in cancer screening will increase, optimizing diagnostic precision, reducing errors, and enhancing workflow efficiency.

3.1.2 AI in tumor classification and staging

AI has advanced tumor classification and staging in medical imaging, with deep learning models accurately analyzing images to differentiate tumor types and aid precise staging for better treatment decisions [29]. In lung cancer, AI analyzes CT images to assess tumor size, morphology, and its relation to surrounding tissues, improving TNM staging and metastasis risk evaluation [24]. Youssefirizi et al. demonstrated AI's effectiveness in bladder cancer, analyzing tumor morphology and invasiveness for better clinical support [30]. Ramesh et al. [31] highlighted AI's role in liver cancer staging through CT-MRI analysis, improving classification accuracy and predicting tumor aggressiveness [31]. This integration aids early detection, particularly for hepatocellular carcinoma, often diagnosed late. AI also improves prostate cancer staging, using MRI and ultrasound. Hassan et al. [32] found AI deep learning models effective in detecting early changes, improving staging and preoperative assessment [32]. As AI evolves, it will further

enhance tumor classification, staging, and personalized treatment planning, boosting diagnostic accuracy and supporting precision medicine.

3.1.3 AI-assisted cancer screening systems

AI-powered cancer screening systems are crucial for early detection, efficiently processing large imaging datasets to detect abnormal patterns and enhance accuracy [33]. In cervical cancer, Poudel et al. [34] showed that AI-driven systems accurately identify features, even in low-quality images, improving accessibility in resource-limited areas [34]. Ng et al. [35] demonstrated AI's effectiveness in breast cancer screening, with deep learning models integrated with mammography to boost lesion detection and diagnostic efficiency [35]. AI also aids in oral cancer screening. Talwar et al. [36] found mobile-based AI tools detect early oral malignancies, improving diagnosis in remote areas [36]. In lung cancer, Zhang et al. [37] noted AI-assisted CT analysis improves nodule detection and malignancy risk assessment, optimizing large-scale screening [37]. As AI advances, cancer screening systems will evolve, refining diagnostic precision, reducing false positives, and enhancing workflow, with broader clinical adoption transforming early detection strategies.

3.2 AI in tumor treatment via medical imaging

AI technology is increasingly playing a pivotal role in cancer treatment. It enhances the effectiveness of radiotherapy, targeted therapy, and immunotherapy by improving accuracy, efficacy prediction, and patient safety. Through the analysis of medical images, clinical data, and biomarkers, AI can identify tumor characteristics, predict treatment efficacy and potential risk factors, and assist physicians in adjusting treatment plans to achieve personalized medicine [38]. In radiotherapy, AI facilitates automated segmentation and dose prediction, enabling precise delineation of tumors and organs at risk while minimizing damage to healthy tissues [39]. AI can also perform rapid dose prediction based on daily imaging, thereby enhancing the efficiency of adaptive radiotherapy. Additionally, AI technology plays a crucial role in targeted and immunotherapy. By integrating imaging and molecular data, it provides more personalized and effective treatment strategies for patients. Furthermore, these technologies contribute to predicting responses to immune checkpoint inhibitors, facilitating the development of personalized immunotherapy approaches [40].

Overall, AI advances precision and personalization in cancer treatment, expanding its role in oncology to improve efficacy, reduce side effects, and enhance patient outcomes.

3.2.1 AI-assisted radiation therapy planning and evaluation

AI techniques have significantly improved the efficiency and quality of radiation therapy by automatically contouring target volumes and organs at risk, as well as predicting dose distribution. By analyzing patient imaging data (e.g., CT, MRI, PET), AI minimizes human error, reduces the risk of adverse reactions, and shifts radiation therapy to a data-driven, precision medicine approach [5]. In adaptive radiotherapy (ART), AI is instrumental in streamlining the replanning process by analyzing daily imaging data to detect anatomical changes. Although current clinical ART systems can meet planning coverage requirements, AI enhances efficiency and supports real-time plan adaptation, particularly in complex cases [41]. In breast cancer, AI optimizes beam trajectories,

adapts target delineation, and adjusts doses for anatomical changes, improving accuracy and reducing normal tissue exposure [42]. AI also aids in radiotherapy evaluation, using radiomics to assess tumor features for efficacy and prognosis prediction [38]. It integrates clinical data to predict treatment outcomes, recurrence, or metastasis [7]. In IMRT and VMAT, AI optimizes dose distribution, ensuring safety and effectiveness. AI reduces clinician workload, enhances dose precision, minimizes healthy tissue exposure, and improves tumor control rates [43].

As AI evolves, its role in radiation therapy will expand, refining precision, fostering collaboration, and improving post-treatment evaluations in intelligent oncology. A recent multi-institutional clinical study evaluated deep learning dose prediction in 622 patients across various cancer types (e.g., breast, cervical, nasopharyngeal) using U-Net based models. Over 53% of AI-generated plans were deemed clinically acceptable, and combination model strategies improved that to 62.6%, demonstrating real-world feasibility of AI-assisted dose planning [44]. Reinforcement learning has also entered clinical research: a recent Phys Med Biol study described an RL-driven framework that mimics human planner decisions to generate head-and-neck simultaneous integrated boost (SIB) plans, aligning with clinical preferences and delivering flexible, efficient planning solutions [45]. While directly generating executable plans using AI raises clinical safety concerns, techniques such as deep learning for dose prediction and reinforcement learning for plan optimization are actively researched to assist clinical physicists and enhance planning workflows. As AI technologies continue to evolve, their role in radiation therapy will expand, improving collaborative planning processes and supporting posttreatment evaluation in intelligent oncology.

3.2.2 AI in interventional therapy and surgery

The rapid development of artificial intelligence (AI) is transforming interventional therapy and surgical procedures. By integrating medical imaging, surgical robotics, deep learning algorithms, and augmented reality (AR), AI enhances precision medicine, optimizes surgical workflows, and improves both treatment safety and success rates [46]. As deep learning, AR, and robotics continue to advance, AI's role in interventional therapy and surgery will expand, providing stronger support for personalized treatments and precise surgical interventions.

3.2.2.1 AI in interventional therapy Interventional therapy, a minimally invasive approach guided by imaging, includes procedures like angioplasty, tumor ablation, and biopsies. AI applications focus on preoperative planning, intraoperative navigation, and postoperative evaluation. In preoperative planning, AI analyzes CT, MRI, and ultrasound data to assess lesions and optimize surgical pathways, improving accuracy [47]. In radio-frequency ablation (RFA), AI predicts tumor-to-vessel distances, enhancing safety. In vascular procedures, AI detects stenosis and optimizes stent placement, reducing procedure time and complications [48]. During surgery, AI guides real-time imaging, aiding precise adjustments and enhancing safety [49]. In vascular interventions, AI improves catheter placement and minimizes radiation. AI-powered navigation systems with AR overlay preoperative imaging onto surgical views for better guidance [50].

Postoperatively, AI analyzes follow-up scans to assess recovery, treatment efficacy, and recurrence risk, such as detecting lung cancer recurrence in follow-up CTs for personalized monitoring [51].

3.2.2.2 AI in surgical applications AI enhances surgery through robotic assistance, intraoperative imaging, and postoperative complication prediction. AI-powered systems like the Da Vinci robot use computer vision and machine learning for high-precision procedures, improving outcomes in prostatectomy [52]. AI also learns from experienced surgeons to refine robotic decision-making in neurosurgery and spinal surgery. Intraoperatively, AI integrates AR and XR to overlay real-time imaging, enhancing tumor resection accuracy. For example, in laparoscopic surgery, AI combines MRI or CT images with surgical views for precise tumor removal [53].

Postoperatively, AI uses deep learning to analyze preoperative, intraoperative, and postoperative data, predicting complications. In prostatectomy, it forecasts risks like urinary incontinence and erectile dysfunction, enabling proactive management for improved recovery [54].

3.2.3 AI in targeted therapy and personalized cancer treatment

Advancements in AI are revolutionizing targeted therapy and personalized cancer treatment by leveraging genetic, proteomic, imaging, and clinical data to tailor optimal therapeutic strategies. AI enhances drug discovery, optimizes treatment regimens, and predicts therapeutic responses, significantly improving precision while minimizing adverse effects [55]. AI-driven approaches enable more precise, efficient, and lower-risk cancer therapies, pushing treatment towards higher accuracy and reduced toxicity [56].

3.2.3.1 AI in targeted therapy for cancer AI plays a key role in targeted therapy by identifying cancer targets, accelerating drug development, and optimizing personalized treatment. Using deep learning, AI analyzes genomic and proteomic data to discover novel cancer targets and predict drug mechanisms. For example, AI identifies genes linked to cancer progression, providing precise therapy targets [57].

AI speeds drug development through molecular docking and virtual screening, evaluating millions of compounds to predict binding affinity and reduce discovery time [58]. In breast cancer, AI has been used to screen HER2-targeted drugs and predict patient responses [59]. AI also combines multi-modal data (genomics, radiomics, and clinical records) to optimize individualized therapy. In non-small cell lung cancer (NSCLC), AI integrates PET/CT with EGFR mutation data to predict drug responses, improving treatment efficacy and minimizing resistance [56]. Additionally, AI identifies mutations like EGFR, ALK, and ROS1, recommending targeted therapies such as Osimertinib or Crizotinib [60].

These findings suggest a growing need for integrative AI frameworks that not only predict drug responses but also account for real-time molecular dynamics. In our view, future development should focus on building closed-loop AI systems that dynamically adapt therapy recommendations based on patient-specific genomic evolution during treatment, enhancing responsiveness and mitigating resistance risk.

3.2.3.2 AI in personalized cancer treatment AI is advancing precision oncology by enabling personalized treatment plans based on individual patient characteristics, enhancing efficacy and minimizing side effects.

AI integrates multi-modal data—genomics, proteomics, radiomics, and clinical data—to optimize personalized cancer treatment models. For example, AI combines MRI, CT, and PET imaging to predict tumor progression and assess responses to radiation, targeted therapy, or immunotherapy [61]. It dynamically adjusts treatment plans by analyzing real-time imaging and clinical data [57].

AI optimizes drug selection through pharmacogenomics, tailoring medications to a patient's genetic profile. For example, AI analyzes CYP450 gene polymorphisms to predict metabolic responses to anticancer drugs, enabling dose adjustments to reduce side effects and enhance efficacy [62].

AI also facilitates real-time treatment response assessments and adjustments. In prostate cancer therapy, AI integrates MRI imaging and PSA levels to track tumor evolution, predict resistance risk, and adjust treatment plans [56, 63].

Based on the reviewed literature and current clinical limitations, we propose a modular AI architecture integrating pharmacogenomic predictors, imaging-based tumor monitoring, and adaptive clinical feedback loops. This structure could serve as a blueprint for real-time personalized oncology systems that continuously refine therapeutic regimens in response to patient-specific trajectories.

3.2.4 Treatment monitoring and adjustment

Continuous monitoring and real-time adjustment of cancer treatment responses are crucial for optimizing therapeutic efficacy, reducing side effects, and personalizing treatment strategies. AI plays a significant role in this domain through real-time imaging analysis, treatment plan modification, individualized prognosis prediction, and remote patient management, significantly enhancing the precision and flexibility of cancer care [24, 64].

3.2.4.1 AI in treatment monitoring AI enables real-time analysis of medical imaging, biomarkers, and clinical data to track therapeutic responses and dynamically adjust treatment plans for optimal outcomes. AI-driven monitoring can detect tumor growth, shrinkage, or metastasis in real-time. For instance, AI integrates CT, MRI, and PET imaging to automatically assess tumor volume changes, providing rapid treatment evaluations and prognostic insights [65].

Beyond response tracking, AI can adjust radiation dosage or modify treatment strategies based on tumor morphology and volume changes. In radiotherapy, AI optimizes radiation dosing dynamically, ensuring precision while minimizing damage to healthy tissues [66]. Additionally, AI enhances personalized treatment decision-making by integrating genomics, radiomics, and clinical data to predict patient resistance to specific therapies and suggest alternative treatments. In liver cancer management, AI combines blood biomarkers with imaging data to forecast patient responses to targeted therapies, enabling proactive treatment modifications [67].

While AI applications in treatment monitoring are rapidly evolving, existing tools often lack cross-modality integration and validation in diverse clinical environments. We suggest that future platforms prioritize interoperability and federated learning to

unify data from imaging, biomarkers, and wearable devices across institutions, thereby improving longitudinal care and model generalizability.

3.2.4.2 AI in remote treatment monitoring With advancements in telemedicine and digital health technologies, AI facilitates remote monitoring of cancer patients, providing intelligent health management solutions. By integrating AI with smart medical devices and wearable sensors, real-time physiological monitoring is possible, tracking parameters such as temperature, blood pressure, glucose levels, and biomarker fluctuations. AI-powered medical sensors can automatically assess a cancer patient's immune status and predict infection risks [68].

AI also bridges healthcare gaps in resource-limited regions by enabling timely medical interventions. AI-driven clinical decision support systems (CDSS) analyze electronic health records (EHR) to identify high-risk patients, optimize follow-up protocols, and lower recurrence rates [69]. Additionally, AI monitors real-time patient feedback, dynamically adjusting treatment parameters to mitigate adverse effects and enhance therapeutic outcomes [70]. To advance AI-supported remote oncology care, we recommend the establishment of standardized AI-telehealth protocols tailored for cancer management. Such protocols should define minimum data fidelity, real-time response mechanisms, and patient risk triage criteria, especially for rural or under-resourced settings. Moreover, incorporating patient-reported outcomes into AI feedback loops could enhance patient-centered care.

4 Challenges in AI-based tumor imaging diagnosis and treatment

AI has significantly advanced diagnostic accuracy, personalized treatment, and patient outcomes, but its clinical implementation faces challenges including data quality, model generalizability, ethical issues, clinical acceptance, and integration with healthcare systems [71]. AI models require high-quality data, but data heterogeneity, inconsistent annotations, and privacy regulations limit their applicability across institutions and devices. Additionally, AI's "black box" nature reduces interpretability, making it hard for clinicians to trust its recommendations [24]. Ethical concerns, such as data privacy, legal responsibility, and algorithmic bias, also hinder widespread adoption [72].

Future efforts should focus on developing large-scale, multi-center, high-quality imaging databases to enhance AI model generalizability. Explainable AI (XAI) should be explored to increase clinician confidence, and robust legal and ethical frameworks must be established to ensure the safe and compliant use of AI in healthcare (Table 1).

4.1 Data quality and standardization issues

AI applications in medical imaging heavily depend on large, high-quality datasets. However, data quality variations, lack of standardization, imaging discrepancies across institutions, and legal restrictions on data sharing pose challenges to AI model generalizability in clinical settings [73].

4.1.1 Heterogeneity and standardization challenges in imaging data

Medical imaging data originate from various hospitals, imaging devices, and acquisition parameters, leading to significant heterogeneity. For example, CT scans of the same lung cancer patient obtained from different scanner brands (e.g., Siemens, GE, Philips) may

Table 1 Application and challenges of AI imaging in tumor diagnosis and treatment

AI application	Function description	Imaging technologies & data sources	Key challenges	Reference
Early tumor detection	AI utilizes deep learning for automated cancer screening, enhancing early detection accuracy	Mammography, CT, MRI, PET; Radiology department databases (PACS), TCIA, UK Biobank	Inconsistent imaging data quality may lead to false positives/negatives	[25]
Tumor classification & staging	AI analyzes imaging features to accurately determine tumor stage, improving diagnostic consistency	MRI, PET-CT, CT; Multi-center clinical imaging databases	Limited generalizability of AI models across different populations and datasets	[24]
Radiotherapy planning	AI automatically delineates tumor targets and optimizes radiation dose distribution	CT, MRI, PET-CT; Radiotherapy planning databases (RT-PACS)	AI decision-making lacks interpretability, reducing clinicians' trust	[41]
Surgical & interventional therapy assistance	AI integrates real-time imaging guidance to enhance surgical precision and minimize errors	Intraoperative CT, real-time MRI, ultrasound imaging, AR-based visualization; Intraoperative imaging databases	High implementation costs and integration challenges within hospital systems	[50]
Targeted & personalized therapy	AI combines radiomics and genomic data to optimize individualized treatment strategies	PET-CT, MRI, Next-Generation Sequencing (NGS), Radiomics databases	Complex integration of imaging and biomarker data	[40]
Treatment monitoring & assessment	AI automatically evaluates imaging data to predict treatment response and disease progression	MRI, CT, PET; Electronic Health Records (EHR), longitudinal follow-up data	Existing AI-based monitoring systems lack long-term clinical validation	[42]

exhibit variations in grayscale values and contrast, impacting AI model performance across datasets [74]. Majumder et al. demonstrated that discrepancies in imaging protocols led to a reduction in AUC by over 15% on external validation cohorts, highlighting the critical need for multicenter validation to ensure model robustness and generalizability [75].

Although the DICOM (Digital Imaging and Communications in Medicine) format is the global standard for medical imaging storage, its implementation varies across institutions. Differences in image post-processing, compression methods, and storage formats complicate cross-hospital and cross-regional data sharing [76]. Without consistent data standardization, AI models may learn biased representations, reducing their reliability and clinical applicability. Seoni et al. [77] pointed out that many multicenter AI imaging studies fail to implement unified normalization or standardization pipelines during training, which is a key factor in the failure of model transferability [77].

To address these challenges, a variety of harmonization techniques have been developed. The ComBat algorithm has been widely adopted to adjust for scanner- and site-specific batch effects. For example, Da-Ano et al. applied a modified ComBat approach to radiotherapy datasets from multiple centers, significantly reducing device-induced bias and enhancing model reproducibility and robustness [78]. In parallel, Guo et al. proposed domain adaptation strategies that align latent feature distributions across domains using deep neural networks, effectively mitigating data drift across time and institutions [79].

Clinically, several AI initiatives have failed due to unresolved imaging heterogeneity. One notable example is the AI-RANO project in neuro-oncology, where a multimodal MRI-based model for glioma classification, trained on data from a single vendor, underperformed on multicenter datasets and was ultimately denied regulatory approval for

clinical integration. This case underscores the necessity of addressing data heterogeneity and conducting cross-domain validation early in the model development lifecycle [80].

4.1.2 Impact of data quality on AI model performance

AI models rely on accurately labeled data, but inconsistencies in medical annotations can introduce bias. Studies indicate that inter-observer variability among radiologists can reach 25% in mammography interpretation, potentially affecting AI model training [81].

Additionally, medical imaging datasets may contain noise, including motion artifacts, metal-induced distortions, and low-resolution images, impacting AI learning processes. Training datasets predominantly sourced from Western populations may also introduce biases, limiting model effectiveness for underrepresented ethnic groups [82].

Furthermore, recent analyses demonstrate that lack of population diversity in datasets exacerbates bias, especially in oncology applications. Delgado-López et al. highlight that AI systems trained predominantly on Western data risk systemic underperformance in underrepresented ethnic populations, undermining diagnostic equity [83].

4.1.3 Challenges in imaging data sharing and privacy protection

Medical imaging data often contain sensitive patient information, necessitating strict compliance with regulations such as the General Data Protection Regulation (GDPR) in Europe and the Health Insurance Portability and Accountability Act (HIPAA) in the United States. These regulations restrict data exchange between hospitals and institutions, limiting AI models' access to multi-center, large-scale datasets required for effective training [84].

To address data-sharing constraints, a novel AI training approach known as Federated Learning has emerged. This method allows models to be trained locally at different institutions without transferring patient data. For instance, Google's federated learning framework has been successfully applied in multi-center breast cancer screening research, demonstrating that AI models can be efficiently trained without the need for direct data sharing [85].

4.1.4 Future directions for addressing data quality and standardization issues

Data quality and standardization challenges hinder AI-driven tumor imaging, with issues like heterogeneous data sources, inconsistent annotations, regulatory restrictions, and privacy concerns affecting model generalizability and clinical use.

Future efforts should focus on developing standardized multi-center databases, such as TCIA and UK Biobank, to enhance AI model robustness [86]. Implementing multi-expert consensus labeling and AI-assisted annotation can reduce bias and improve accuracy. Privacy-preserving techniques like differential privacy, homomorphic encryption, and federated learning can enable secure data sharing [87]. Moreover, AI-based pre-processing methods—such as deep-learning denoising (e.g., deep convolutional neural networks for CT/MRI denoising) and intensity normalization pipelines—have demonstrated significant improvements in image quality before model training, reducing noise and standardizing contrast [88, 89]. These approaches directly support downstream model performance and clinical adoption.

4.2 Transparency and explainability in AI models

The rapid adoption of AI in medical imaging has significantly improved cancer diagnosis and treatment planning. However, the “black box” nature of AI models poses challenges in clinical trust and transparency. Physicians and patients need to comprehend AI-generated decisions to ensure safety and reliability [90]. The emergence of Explainable AI (XAI) aims to enhance AI transparency, enabling physicians to validate diagnostic results and mitigate risks in medical decision-making [91].

4.2.1 The “black box” problem and the need for transparency in AI models

Modern AI models, particularly deep neural networks (DNNs) and convolutional neural networks (CNNs), exhibit highly complex architectures that make their decision-making processes difficult to interpret. For instance, in breast cancer diagnosis, AI models can analyze mammograms or MRI scans through deep learning, yet they do not provide explicit pathological reasoning, making it challenging for physicians to fully trust their conclusions [92]. Viswanathan and Parmar emphasize that interpretability is not merely a technical goal but a regulatory imperative, especially as opaque models may propagate hidden biases that are difficult to detect post-deployment [93]. This underscores the urgent need for XAI frameworks integrated at the development stage.

Lack of transparency affects AI adoption in clinical settings. Studies indicate that physicians are more likely to trust AI systems that offer visual explanations (e.g., heatmaps and feature importance analyses) rather than those that provide only final diagnostic predictions [94]. Enhancing AI transparency not only builds physician confidence but also facilitates broader AI implementation across diverse healthcare institutions.

4.2.2 Explainable AI (XAI) in medical imaging

AI can enhance transparency through visualization techniques such as Grad-CAM, LIME, and SHAP, which illustrate how AI models make diagnostic decisions. For example, in lung cancer CT analysis, Grad-CAM generates heatmaps that highlight the most relevant lesion areas identified by the AI model, aiding radiologists in understanding its diagnostic rationale [95].

Beyond visualization, XAI also employs rule-based approaches, such as decision trees and logistic regression, to provide explicit diagnostic reasoning. For instance, rule-based AI models can assess patient age, tumor size, and molecular features to predict breast cancer recurrence risk, offering a clear decision pathway [96].

However, studies have shown that post-hoc explanation methods like Grad-CAM or SHAP may not always reflect true model reasoning and can be susceptible to noise, potentially misleading clinicians. Reyes et al. stress the importance of explanation fidelity to ensure that visualizations correspond to valid model logic [97]. To address this, hybrid models are being explored, integrating interpretable layers or attention mechanisms within deep networks. Viswanathan and Parmar propose combining black-box AI with interpretable modules to maintain performance while offering clinical auditability [93]. Concept-based XAI further bridges the gap by linking predictions to medical features such as tumor location or histology, enabling clinicians to evaluate AI decisions through familiar semantic dimensions [98].

4.2.3 The role of transparent AI in medical regulation and ethics

Enhancing AI transparency is crucial for regulatory bodies like the FDA, which are advancing explainability standards to ensure fairness and safety. Research shows that explainable AI can reduce diagnostic errors due to bias and improve fairness across demographics [99].

Transparency also builds patient trust. When patients understand how AI reaches a diagnosis, they are more likely to accept AI-assisted treatment recommendations. For example, in skin cancer screening, AI can visually demonstrate lesion detection with explanations, making diagnoses clearer to patients [100].

Looking ahead, XAI can integrate imaging, clinical, and genomic data to improve AI interpretability in personalized medicine. Strengthening human-AI interaction with real-time physician queries can refine clinical decisions. Additionally, establishing legal and ethical frameworks for AI transparency will ensure compliance with medical ethics and regulations, advancing precision medicine through multi-modal AI, visualization, and standardized regulations. Transparency is also pivotal for ensuring regulatory compliance. Reyes et al. argue that explainable models should form the backbone of regulatory submission, as opacity impedes risk–benefit assessments by oversight bodies such as the FDA and EMA [97].

4.3 Barriers to clinical implementation

Despite AI's transformative potential in medical imaging and oncology, its widespread clinical adoption faces multiple barriers, including technical limitations, workflow integration, regulatory and ethical challenges, cost-effectiveness concerns, and physician–patient acceptance [101].

4.3.1 Technical challenges

AI models lack standardized data quality benchmarks and face generalization issues across institutions and regions. Most AI models are trained on datasets from specific hospitals or research centers, which may not fully represent global patient populations. Studies have shown that AI models trained on Western patient data perform well in radiology but exhibit reduced accuracy when applied to Asian or African populations, highlighting the generalization challenge [102].

Another critical challenge in AI clinical adoption is its decision-making opacity. Most deep learning models operate as “black boxes,” making it difficult for physicians to interpret their recommendations. For instance, in radiotherapy planning, AI might propose an optimal dose distribution but fail to provide a transparent computational rationale, reducing physician confidence in AI-assisted treatment planning [103]. The challenge of real-world generalizability is further demonstrated in large-scale assessments. Yang et al. showed that AI models trained under controlled conditions often fail in clinical generalization, even when demographic parity is maintained, revealing hidden performance variability across populations [98].

4.3.2 Integration into medical workflow

Compatibility with existing systems remains a challenge, as Radiology Information Systems (RIS) and Picture Archiving and Communication Systems (PACS) are often provided by different vendors, leading to discrepancies in data formats and communication

protocols. These inconsistencies make seamless AI integration difficult [104]. For example, varying imaging storage standards across healthcare institutions complicate AI model deployment for multi-center data analysis.

A major concern in clinical practice is that while AI enhances diagnostic efficiency, it may also increase workload. AI-generated alerts for “suspicious” cases often require further review by radiologists, potentially increasing rather than decreasing their workload [105].

4.3.3 Regulatory and ethical issues

AI adoption in healthcare is subject to strict regulations, such as the General Data Protection Regulation (GDPR) in Europe and the Health Insurance Portability and Accountability Act (HIPAA) in the United States. These regulations limit cross-institutional data sharing, restricting AI applications in multi-center environments [106]. Legal responsibility for AI-driven misdiagnoses remains unresolved—should liability fall on physicians, hospitals, or AI developers? Current legal frameworks lack clarity on AI accountability in medical decision-making, particularly in high-risk fields such as radiotherapy and pathology [107]. For example, in the widely publicized Watson for Oncology project, AI-generated treatment recommendations were found to be clinically inappropriate in some cases. Although no formal legal action was taken, the incident highlighted ambiguities in determining whether responsibility lies with the physician, the hospital, or the algorithm developer [108]. Under GDPR Article 22, fully automated decision-making that significantly affects individuals—such as AI-based diagnoses—requires specific legal safeguards, including the right to human intervention. Gerke et al. argue that if biased training data leads to discriminatory outcomes (e.g., by race or gender), clinicians may be held accountable for deploying such systems, even if the discrimination was algorithmic in origin [109].

In terms of regulatory oversight, the FDA has proposed a framework for “adaptive AI/ML-based software” with real-time monitoring, yet liability guidelines for clinical harm remain vague. Meanwhile, the EU’s forthcoming AI Act classifies medical AI as “high-risk,” mandating requirements for algorithm transparency, traceability, and post-market surveillance [110]. These policies aim to establish shared accountability across stakeholders, but gaps remain in legal practice.

Thus, the ethical imperative in deploying AI extends beyond technical validation—it demands transparent governance, clear human oversight, and equitable allocation of responsibility. These concerns are especially critical in high-stakes fields such as radiotherapy and pathology, where erroneous AI decisions could have life-altering consequences.

4.3.4 Cost-effectiveness analysis

The implementation of AI in healthcare entails significant costs, including algorithm development, hardware infrastructure, data storage, and maintenance. AI applications in radiology require substantial computational resources, which may exceed the budgets of smaller hospitals [71].

Although AI has the potential to reduce misdiagnoses and unnecessary tests, its long-term economic benefits remain uncertain. For instance, while AI improves early breast

cancer detection rates, its impact on reducing overall treatment costs still requires long-term follow-up studies [111].

4.3.5 Acceptance by clinicians and patients

Clinicians' trust in AI significantly influences its clinical adoption. A survey of radiologists found that while most recognize AI's potential, fewer than 50% would rely solely on AI-generated diagnoses without human verification [112]. This highlights the need for enhanced AI explainability to increase clinician confidence.

Patient acceptance also affects AI's success in healthcare. A study on prostate cancer patients revealed that although AI improves diagnostic accuracy, many patients prefer physician intuition over AI-based analysis [112].

4.4 Regulatory and ethical challenges

AI's widespread deployment in medical imaging and oncology is constrained by regulatory and ethical concerns, including model transparency, data privacy, legal liability, and fairness. Global regulatory agencies and ethicists emphasize the need for secure, compliant, and unbiased AI applications in healthcare [113]. Ensuring AI safety, compliance, and fairness requires collaboration among governments, regulators, hospitals, and developers [114].

4.4.1 Current AI healthcare regulations

Governments worldwide are establishing legal frameworks for AI in medicine. The U.S. Food and Drug Administration (FDA) has approved multiple AI-driven medical imaging tools and is developing new regulatory guidelines to ensure safety and efficacy [115]. The European Union's AI Act focuses on risk management and mandates strict data transparency for medical AI systems.

Despite evolving AI regulations, oversight remains challenging. Traditional medical device approval processes apply to static products, whereas AI models continually learn and adapt, posing challenges for existing regulatory frameworks [116].

4.4.2 Ethical issues in medical imaging AI

Data privacy is a key ethical concern. AI models require large medical imaging datasets containing sensitive patient information, making data sharing and privacy protection a challenge [117]. The GDPR requires patient consent for data use, limiting AI model training in multi-center studies.

Bias in AI training data can lead to diagnostic disparities, as many models are trained on Western datasets, affecting performance in diverse ethnic groups [118]. To ensure fairness, AI systems must be audited independently and tested on multi-ethnic, multi-regional datasets. Legal liability for AI-driven medical errors is unclear. If AI contributes to errors, should responsibility lie with physicians, hospitals, AI developers, or the system itself? Currently, AI serves as a decision-support tool, with physicians making final decisions [119]. As AI advances, the question of its independent legal responsibility remains debated.

4.4.3 Future directions

To ensure safe, compliant, and fair AI application in medical imaging and oncology, several steps are needed. First, a globally unified regulatory framework will facilitate international collaboration, addressing current regulatory discrepancies [120]. Second, advancing explainable AI (XAI) will enhance transparency, improving trust among physicians and patients [121]. Lastly, AI systems should undergo thorough ethical review to prevent bias before clinical deployment [72].

In summary, AI's widespread use in medical imaging and oncology requires careful attention to regulatory and ethical issues. Governments are establishing systems to ensure AI's safety, fairness, and transparency. Future efforts should refine regulatory frameworks, develop XAI models, and enhance data privacy protections to ensure compliant AI application.

5 Conclusion

5.1 The potential of AI in tumor imaging

AI has demonstrated immense potential in tumor imaging, significantly advancing early cancer detection, personalized treatment planning, and treatment response monitoring. In recent years, AI has been widely applied in imaging analysis for various cancers, including breast, lung, and prostate cancer, improving diagnostic accuracy and efficiency [114]. Additionally, AI's role in oncology treatment is expanding, particularly in optimizing radiotherapy planning, real-time image-guided therapy, and individualized prognosis prediction [24]. As AI algorithms continue to evolve and multi-modal imaging data integration progresses, AI is expected to further enhance the precision of tumor diagnostics and drive personalized medicine forward. For example, AI can integrate radiomics, genomics, and clinical data to develop more accurate cancer prediction models, guiding individualized treatment strategies [122].

5.2 Future directions and research areas

Despite AI's impact on tumor imaging, challenges persist. Many AI models are trained on single-center or specific ethnic datasets, limiting generalizability. Future research should focus on optimizing AI algorithms and expanding multi-center datasets for broader applicability [123]. Interdisciplinary collaboration and multi-modal data integration are essential for advancing oncology imaging. AI development requires synergy between computer science, medical imaging, pathology, and oncology, with exploration of AI in PET/MRI imaging combined with genomic data to enhance tumor biology understanding [124]. Clinical validation is crucial, and prospective, multi-center trials are needed to evaluate AI's real-world efficacy and safety [125].

Despite enthusiasm for open data, practical collaboration barriers remain: stringent privacy regulations (e.g., GDPR, HIPAA), disparate institutional workflows (e.g., pre-processing pipelines, DICOM conventions), and technical incompatibilities (e.g., RIS/PACS vendor lock-in). Future efforts should promote standard operating procedures, regulatory harmonization, and interoperable data frameworks to enable effective international AI studies.

5.3 Limitations and outlook

Despite progress, AI-driven tumor imaging faces challenges. Data quality and annotation consistency are key issues, as variations in imaging protocols and radiologists' annotations affect model performance and reliability [33]. Additionally, AI's transparency and clinical interpretability need improvement. Many AI models lack interpretability, hindering physician understanding of AI-generated diagnoses. Future research should focus on more interpretable models, such as decision trees or deep learning with integrated visualization, to enhance transparency and trust [126].

Ethical and legal issues remain unresolved, including privacy protection, data-sharing regulations, and liability for AI-related diagnostic errors. Legal accountability for errors is unclear—should responsibility fall on physicians, hospitals, or AI developers? Future research should develop regulatory frameworks to ensure compliant AI deployment in healthcare [127].

We advocate that explainable AI (XAI) and robust regulatory frameworks are no longer optional but foundational. XAI enhances clinician trust, enables regulatory review (FDA, EU AI Act), and helps detect bias. We also call for the development of adaptive regulation—combining premarket risk assessment with post-market monitoring—to ensure safety and fairness as AI systems evolve.

6 Conclusion

AI is rapidly transforming tumor imaging, significantly enhancing cancer diagnosis, treatment planning, and prognosis prediction. Future research should focus on optimizing AI algorithms, fostering interdisciplinary collaboration, integrating multi-modal data, and conducting extensive clinical validation to enhance AI's real-world applicability. Additionally, addressing data quality, transparency, ethical, and regulatory concerns is crucial to ensuring AI's safe, efficient, and equitable integration into oncology imaging.

Author contributions

JYH, YZX contributed to the conception of the study. SQG and LRW performed the data analyses and wrote the manuscript. JYH and JJZ contributed constructive discussions. JYY and DY collaborated to write and revise the article.

Funding

This work was supported by grants from 2024 Ningbo Public Welfare Science and Technology Plan Key Project (No. 2024S032), Medical and Health Research Project of Zhejiang Provinc (2024KY1481, 2025KY1296).

Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

Not applicable.

Human and animal rights

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 26 April 2025 / Accepted: 25 July 2025

Published online: 26 August 2025

References

1. Marti-Bonmati L, et al. Considerations for artificial intelligence clinical impact in oncologic imaging: an AI4HI position paper. *Insights Imaging*. 2022;13(1):89. <https://doi.org/10.1186/s13244-022-01220-9>.
2. Forghani R. Precision digital oncology: emerging role of radiomics-based biomarkers and artificial intelligence for advanced imaging and characterization of brain tumors. *Radiol Imaging Cancer*. 2020;2(4): e190047. <https://doi.org/10.1148/rycan.2020190047>.
3. Coccia M. Deep learning technology for improving cancer care in society: new directions in cancer imaging driven by artificial intelligence. *Technol Soc*. 2020;60:101198. <https://doi.org/10.1016/j.techsoc.2019.101198>.
4. Bonmati LM, et al. CHAIMELEON project: creation of a Pan-European repository of health imaging data for the development of AI-powered cancer management tools. *Front Oncol*. 2022. <https://doi.org/10.3389/fonc.2022.742701>.
5. van de Sande D, et al. Artificial intelligence based treatment planning of radiotherapy for locally advanced breast cancer. *Phys Imaging Radiat Oncol*. 2021;20:111–6. <https://doi.org/10.1016/j.phro.2021.11.007>.
6. Castiglioni I, et al. AI applications to medical images: from machine learning to deep learning. *Phys Med*. 2021;83:9–24. <https://doi.org/10.1016/j.ejmp.2021.02.006>.
7. Tadiboina SN. The use of AI in advanced medical imaging. *J Posit School Psychol*. 2022;6(11):1939–46.
8. Zhovannik I, et al. Radiomics integration into a picture archiving and communication system. *Phys Imaging Radiat Oncol*. 2021;20:30–3. <https://doi.org/10.1016/j.phro.2021.09.007>.
9. Jorg T, et al. A novel reporting workflow for automated integration of artificial intelligence results into structured radiology reports. *Insights Imaging*. 2024;15(1):80. <https://doi.org/10.1186/s13244-024-01660-5>.
10. Shen D, Wu G, Suk H-I. Deep learning in medical image analysis. 2017;19:221–48. <https://doi.org/10.1146/annurev-bioeng-071516-044442>.
11. Litjens G, et al. A survey on deep learning in medical image analysis. *Med Image Anal*. 2017;42:60–88. <https://doi.org/10.1016/j.media.2017.07.005>.
12. Chen X, et al. Recent advances and clinical applications of deep learning in medical image analysis. *Med Image Anal*. 2022;79:102444. <https://doi.org/10.1016/j.media.2022.102444>.
13. Wang J, et al. A review of deep learning on medical image analysis. *Mobile Networks Appl*. 2021;26(1):351–80. <https://doi.org/10.1007/s11036-020-01672-7>.
14. Hu X. Deep learning for medical image analysis. *mediaTUM Gesamtbestand*, 2021; p. 108
15. Azam MA, et al. A review on multimodal medical image fusion: compendious analysis of medical modalities, multimodal databases, fusion techniques and quality metrics. *Comput Biol Med*. 2022;144:105253. <https://doi.org/10.1016/j.compbio.2022.105253>.
16. Huang B, et al. A review of multimodal medical image fusion techniques. *Comput Math Methods Med*. 2020;2020:8279342. <https://doi.org/10.1155/2020/8279342>.
17. Guo Z, et al. deep learning-based image segmentation on multimodal medical imaging. *IEEE Trans Radiat Plasma Med Sci*. 2019;3(2):162–9. <https://doi.org/10.1109/trpms.2018.2890359>.
18. Saleh MA, et al. A brief analysis of multimodal medical image fusion techniques. *Electronics*. 2023;12(1):97.
19. Ullah Khan S, et al. Multimodal medical image fusion towards future research: a review. *J King Saud Univ Comput Inf Sci*. 2023;35(8):101733. <https://doi.org/10.1016/j.jksuci.2023.101733>.
20. Daldrup-Link H. Artificial intelligence applications for pediatric oncology imaging. *Pediatr Radiol*. 2019;49(11):1384–90. <https://doi.org/10.1007/s00247-019-04360-1>.
21. Chakrabarty N, Mahajan A. Imaging analytics using artificial intelligence in oncology: a comprehensive review. *Clin Oncol (R Coll Radiol)*. 2024;36(8):498–513. <https://doi.org/10.1016/j.clon.2023.09.013>.
22. Chen MM, et al. Artificial intelligence in oncologic imaging. *Eur J Radiol Open*. 2022;9:100441. <https://doi.org/10.1016/j.ejro.2022.100441>.
23. Shastry KA, Sanjay HA. Cancer diagnosis using artificial intelligence: a review. *Artif Intell Rev*. 2022;55(4):2641–73. <https://doi.org/10.1007/s10462-021-10074-4>.
24. Bi WL, et al. Artificial intelligence in cancer imaging: clinical challenges and applications. *CA Cancer J Clin*. 2019;69(2):127–57. <https://doi.org/10.3322/caac.21552>.
25. Ijaz MF, Woźniak M. Recent advances in deep learning and medical imaging for cancer treatment. *Cancers*. 2024;16(4):700.
26. Kaur C, Garg U. Artificial intelligence techniques for cancer detection in medical image processing: a review. *Mater Today-Proc*. 2023;81:806–9. <https://doi.org/10.1016/j.matpr.2021.04.241>.
27. Hunter B, Hindocha S, Lee RW. The role of artificial intelligence in early cancer diagnosis. *Cancers*. 2022;14(6):1524.
28. Zheng D, He X, Jing J. Overview of artificial intelligence in breast cancer medical imaging. *J Clin Med*. 2023;12(2):419.
29. Murtaza G, et al. Deep learning-based breast cancer classification through medical imaging modalities: state of the art and research challenges. *Artif Intell Rev*. 2019;53:1655–720.
30. Yousefrizi F, et al. AI-based detection, classification and prediction/prognosis in medical imaging: towards radiophenomics. *PET Clin*. 2022;17(1):183–212. <https://doi.org/10.1016/j.cpet.2021.09.010>.
31. Ramesh P, Karuppasamy R, Veerappapillai S. A review on recent advancements in diagnosis and classification of cancers using artificial intelligence. *Biomedicine*. 2020;10(3):5–17. <https://doi.org/10.37796/2211-8039.1012>.
32. Hassan MR, et al. Prostate cancer classification from ultrasound and MRI images using deep learning based explainable artificial intelligence. *Futur Gener Comput Syst*. 2022;127:462–72. <https://doi.org/10.1016/j.future.2021.09.030>.
33. Wang B, et al. AI-assisted CT imaging analysis for COVID-19 screening: building and deploying a medical AI system. *Appl Soft Comput*. 2021;98:106897. <https://doi.org/10.1016/j.asoc.2020.106897>.
34. Poudel K et al. AI-Assisted Cervical Cancer Screening.
35. Ng AY, et al. Prospective implementation of AI-assisted screen reading to improve early detection of breast cancer. *Nat Med*. 2023;29(12):3044–9. <https://doi.org/10.1038/s41591-023-02625-9>.
36. Talwar V, et al. AI-assisted screening of oral potentially malignant disorders using smartphone-based photographic images. *Cancers*. 2023;15(16):4120.
37. Zhang Y, et al. Lung nodule detectability of artificial intelligence-assisted CT image reading in lung cancer screening. *Curr Med Imaging*. 2022;18(3):327–34. <https://doi.org/10.2174/1573405617666210806125953>.
38. Olawade DB, et al. Integrating AI into cancer immunotherapy-a narrative review of current applications and future directions. *Diseases*. 2025. <https://doi.org/10.3390/diseases13010024>.

39. Peng H, et al. An AI-based approach for modeling the synergy between radiotherapy and immunotherapy. *Sci Rep*. 2024;14(1):8250. <https://doi.org/10.1038/s41598-024-58684-6>.
40. Xu Z, et al. Applying artificial intelligence for cancer immunotherapy. *Acta Pharmaceutica Sinica B*. 2021;11(11):3393–405. <https://doi.org/10.1016/j.apsb.2021.02.007>.
41. Vandewinckele L, et al. Overview of artificial intelligence-based applications in radiotherapy: recommendations for implementation and quality assurance. *Radiother Oncol*. 2020;153:55–66. <https://doi.org/10.1016/j.radonc.2020.09.008>.
42. Sheng Y, et al. Artificial intelligence applications in intensity modulated radiation treatment planning: an overview. *Quant Imaging Med Surg*. 2021;11(12):4859–80. <https://doi.org/10.21037/qims-21-208>.
43. Santoro M, et al. Recent applications of artificial intelligence in radiotherapy: where we are and beyond. *Appl Sci*. 2022;12(7):3223.
44. Hou Z, et al. Deep learning-powered radiotherapy dose prediction: clinical insights from 622 patients across multiple sites tumor at a single institution. *Radiat Oncol*. 2025;20(1):80. <https://doi.org/10.1186/s13014-025-02634-7>.
45. Yang D, et al. Reinforcement learning-driven automated head and neck simultaneous integrated boost (SIB) radiation therapy: flexible treatment planning aligned with clinical preferences. *Phys Med Biol*. 2025. <https://doi.org/10.1088/1361-6560/adcb84>.
46. Geevarghese R, et al. Interventional oncology: 2024 update. *Can Assoc Radiol J*. 2024;75(3):658–70. <https://doi.org/10.1177/08465371241236152>.
47. Glielmo P, et al. Artificial intelligence in interventional radiology: state of the art. *Eur Radiol Exp*. 2024;8(1):62. <https://doi.org/10.1186/s41747-024-00452-2>.
48. Bai Y, et al. Effect of interventional therapy on iliac venous compression syndrome evaluated and diagnosed by artificial intelligence algorithm-based ultrasound images. *J Healthc Eng*. 2021;2021:5755671. <https://doi.org/10.1155/2021/5755671>.
49. Zhang J, et al. How AI and robotics will advance interventional radiology: narrative review and future perspectives. *Diagnostics*. 2024;14(13):1393.
50. Iezzi R, et al. Artificial intelligence in interventional radiology: a literature review and future perspectives. *J Oncol*. 2019;2019:6153041. <https://doi.org/10.1155/2019/6153041>.
51. Boeken T, et al. Artificial intelligence in diagnostic and interventional radiology: Where are we now? *Diagn Interv Imaging*. 2023;104(1):1–5. <https://doi.org/10.1016/j.diii.2022.11.004>.
52. Hashimoto DA, et al. Artificial intelligence in surgery: promises and perils. *Ann Surg*. 2018;268(1):70–6. <https://doi.org/10.1097/sla.0000000000002693>.
53. Qayyum A et al. Can we revitalize interventional healthcare with AI-XR surgical metaverses? 2023. 496–503.
54. Hu J, Chen J, Niu Y. Prospect of the application of interventional therapy in thoracic surgery. *Zhongguo fei ai za zhi = Chin J Lung Cancer*. 2020;23:514–5. <https://doi.org/10.3779/j.issn.1009-3419.2020.101.09>.
55. Schork NJ. Artificial intelligence and personalized medicine. *Cancer Treat Res*. 2019;178:265–83. https://doi.org/10.1007/978-3-030-16391-4_11.
56. Saeed RF, et al. Targeted therapy and personalized medicine. *Cancer Treat Res*. 2023;185:177–205. https://doi.org/10.1007/978-3-031-27156-4_10.
57. Sollini M, et al. Artificial intelligence and hybrid imaging: the best match for personalized medicine in oncology. *Eur J Hybrid Imaging*. 2020;4(1):24. <https://doi.org/10.1186/s41824-020-00094-8>.
58. Blasiak A, Khong J, Kee T. CURATE.AI: Optimizing personalized medicine with artificial intelligence. *SLAS Technol*. 2020;25(2):95–105. <https://doi.org/10.1177/2472630319890316>.
59. Taherdoost H, Ghofrani A. AI's role in revolutionizing personalized medicine by reshaping pharmacogenomics and drug therapy. *Intell Pharm*. 2024;2(5):643–50. <https://doi.org/10.1016/j.ipha.2024.08.005>.
60. Johnson KB, et al. Precision Medicine, AI, and the Future of Personalized Health Care. *Clin Transl Sci*. 2021;14(1):86–93. <https://doi.org/10.1111/cts.12884>.
61. Pub I. AI-driven precision medicine: revolutionizing personalized treatment plans. *Int J Comput Eng Technol*. 2024;15:455–74. <https://doi.org/10.5281/zenodo.13843057>.
62. Khansari N. AI machine learning improves personalized cancer therapies. *Australas Med J*. 2024;17(2):1166–73. <https://doi.org/10.21767/AMJ.2024.4019>.
63. Zahra MA, et al. The synergy of artificial intelligence and personalized medicine for the enhanced diagnosis, treatment, and prevention of disease. *Drug Metab Pers Ther*. 2024;39(2):47–58. <https://doi.org/10.1515/dmpt-2024-0003>.
64. Verlingue L, et al. Artificial intelligence in oncology: ensuring safe and effective integration of language models in clinical practice. *Lancet Reg Health Eur*. 2024;46:101064. <https://doi.org/10.1016/j.lanepe.2024.101064>.
65. Pei Q, et al. Artificial intelligence in clinical applications for lung cancer: diagnosis, treatment and prognosis. *Clin Chem Lab Med*. 2022;60(12):1974–83. <https://doi.org/10.1515/cclm-2022-0291>.
66. Ho D. Artificial intelligence in cancer therapy. *Science*. 2020;367(6481):982–3. <https://doi.org/10.1126/science.aaz3023>.
67. Calderaro J, et al. Artificial intelligence for the prevention and clinical management of hepatocellular carcinoma. *J Hepatol*. 2022;76(6):1348–61. <https://doi.org/10.1016/j.jhep.2022.01.014>.
68. Haick H, Tang N. Artificial intelligence in medical sensors for clinical decisions. *ACS Nano*. 2021;15(3):3557–67. <https://doi.org/10.1021/acsnano.1c00085>.
69. Dangi RR, Sharma A, Vageriya V. Transforming Healthcare in low-resource settings with artificial intelligence: recent developments and outcomes. <https://doi.org/10.1111/phn.13500>
70. Megbawon A, et al. Integrating artificial intelligence in medical imaging for precision therapy: the role of AI in segmentation, laser-guided procedures, and protective shielding. *World J Adv Res Rev*. 2024. <https://doi.org/10.30574/wjarr.2024.23.3.2751>.
71. Cheung HMC, Rubin D. Challenges and opportunities for artificial intelligence in oncological imaging. *Clin Radiol*. 2021;76(10):728–36. <https://doi.org/10.1016/j.crad.2021.03.009>.
72. Khanna S, et al. Ethical challenges arising from the integration of artificial intelligence (AI) in oncological management. *Int J Responsib Artif Intell*. 2020;10(8):34–44.
73. Willemink MJ, et al. Preparing medical imaging data for machine learning. *Radiology*. 2020;295(1):4–15. <https://doi.org/10.1148/radiol.2020192224>.

74. Jeon K, et al. Advancing medical imaging research through standardization: the path to rapid development, rigorous validation, and robust reproducibility. *Investig Radiol.* 2025;60(1):1.
75. Majumder S, et al. State of the art: radiomics and radiomics-related artificial intelligence on the road to clinical translation. *BJR Open.* 2024;6(1):tzad004. <https://doi.org/10.1093/bjro/tzad004>.
76. Kondylakis H, et al. Data infrastructures for AI in medical imaging: a report on the experiences of five EU projects. *Eur Radiol Exp.* 2023;7(1):20. <https://doi.org/10.1186/s41747-023-00336-x>.
77. Seoni S, et al. All you need is data preparation: a systematic review of image harmonization techniques in Multi-center/device studies for medical support systems. *Comput Methods Programs Biomed.* 2024;250:108200. <https://doi.org/10.1016/j.cmpb.2024.108200>.
78. Da-Ano R, et al. Performance comparison of modified ComBat for harmonization of radiomic features for multicenter studies. *Sci Rep.* 2020;10(1):10248. <https://doi.org/10.1038/s41598-020-66110-w>.
79. Guo LL, et al. Evaluation of domain generalization and adaptation on improving model robustness to temporal dataset shift in clinical medicine. *Sci Rep.* 2022;12(1):2726. <https://doi.org/10.1038/s41598-022-06484-1>.
80. Bakas S, et al. Artificial intelligence for response assessment in neuro oncology (AI-RANO), part 2: recommendations for standardisation, validation, and good clinical practice. *Lancet Oncol.* 2024;25(11):e589–601. [https://doi.org/10.1016/S1473-0245\(24\)00315-2](https://doi.org/10.1016/S1473-0245(24)00315-2).
81. Van Ooijen P. Quality and curation of medical images and data: opportunities, applications and Risks. Cham: Springer International Publishing; 2019. p. 247–55.
82. Isgut M, et al. Systematic review of advanced AI methods for improving healthcare data quality in post COVID-19 era. *IEEE Rev Biomed Eng.* 2023;16:53–69. <https://doi.org/10.1109/RBME.2022.3216531>.
83. Delgado-López PD, et al. Artificial intelligence in neuro-oncology: methodological bases, practical applications and ethical and regulatory issues. *Clin Transl Oncol.* 2025. <https://doi.org/10.1007/s12094-025-03948-4>.
84. Panayides AS, et al. AI in medical imaging informatics: current challenges and future directions. *IEEE J Biomed Health Inform.* 2020;24(7):1837–57. <https://doi.org/10.1109/JBHI.2020.2991043>.
85. Prevedello LM, et al. Challenges related to artificial intelligence research in medical imaging and the importance of image analysis competitions. *Radiol Artif Intell.* 2019;1(1): e180031. <https://doi.org/10.1148/ryai.2019180031>.
86. Diaz O, et al. Data preparation for artificial intelligence in medical imaging: a comprehensive guide to open-access platforms and tools. *Physica Med.* 2021;83:25–37. <https://doi.org/10.1016/j.ejmp.2021.02.007>.
87. Hornung AL, et al. Artificial intelligence and spine imaging: limitations, regulatory issues and future direction. *Eur Spine J.* 2022;31(8):2007–21. <https://doi.org/10.1007/s00586-021-07108-4>.
88. Melazzini L, et al. AI for image quality and patient safety in CT and MRI. *Eur Radiol Exp.* 2025;9(1):28. <https://doi.org/10.1186/s41747-025-00562-5>.
89. Huber NR, Missert AD, Erickson BJ. Magician's corner: 7. Using convolutional neural networks to reduce noise in medical images. *Radiol Artif Intell.* 2020;2(5): e200036. <https://doi.org/10.1148/ryai.2020200036>.
90. Chen H, et al. Explainable medical imaging AI needs human-centered design: guidelines and evidence from a systematic review. *NPJ Digit Med.* 2022;5(1):156. <https://doi.org/10.1038/s41746-022-00699-2>.
91. Marey A, et al. Explainability, transparency and black box challenges of AI in radiology: impact on patient care in cardiovascular radiology. *Egypt J Radiol Nucl Med.* 2024;55(1):183. <https://doi.org/10.1186/s43055-024-01356-2>.
92. Liu W, et al. Explainable AI for medical image analysis in medical cyber-physical systems: enhancing transparency and trustworthiness of IoMT. *IEEE J Biomed Health Inf.* 2023. <https://doi.org/10.1109/JBHI.2023.3336721>.
93. Viswanathan VS, Parmar V, Madabhushi A. Towards equitable AI in oncology. *Nat Rev Clin Oncol.* 2024;21(8):628–37. <https://doi.org/10.1038/s41571-024-00909-8>.
94. Garg P, Sharma M, Kumar P. Transparency in diagnosis: unveiling the power of deep learning and explainable AI for medical image interpretation. *Arab J Sci Eng.* 2025. <https://doi.org/10.1007/s13369-024-09896-5>.
95. Salahuddin Z, et al. Transparency of deep neural networks for medical image analysis: a review of interpretability methods. *Comput Biol Med.* 2022;140:105111. <https://doi.org/10.1016/j.combiomed.2021.105111>.
96. Teng Z, et al. A literature review of artificial intelligence (AI) for medical image segmentation: from AI and explainable AI to trustworthy AI. *Quant Imaging Med Surg.* 2024;14(12):9620–52. <https://doi.org/10.21037/qims-24-723>.
97. Reyes M, et al. On the interpretability of artificial intelligence in radiology: challenges and opportunities. *Radiol Artif Intell.* 2020;2(3): e190043. <https://doi.org/10.1148/ryai.2020190043>.
98. Yang Y, et al. The limits of fair medical imaging AI in real-world generalization. *Nat Med.* 2024;30(10):2838–48. <https://doi.org/10.1038/s41591-024-03113-4>.
99. Raghavan K, Balasubramanian S, Veezhinathan K. Explainable artificial intelligence for medical imaging: review and experiments with infrared breast images. *Comput Intell.* 2024;40(3): e12660. <https://doi.org/10.1111/coin.12660>.
100. Roy S, Pal D, Meena T. Explainable artificial intelligence to increase transparency for revolutionizing healthcare ecosystem and the road ahead. *Network Modeling Anal Health Inf Bioinform.* 2023;13(1):4. <https://doi.org/10.1007/s13721-023-00437-y>.
101. Lokaj B, et al. Barriers and facilitators of artificial intelligence conception and implementation for breast imaging diagnosis in clinical practice: a scoping review. *Eur Radiol.* 2024;34(3):2096–109. <https://doi.org/10.1007/s00330-023-10181-6>.
102. Eltawil FA, et al. Analyzing barriers and enablers for the acceptance of artificial intelligence innovations into radiology practice: a scoping review. *Tomography.* 2023;9(4):1443–55.
103. Hindocha S, et al. Artificial intelligence for radiotherapy auto-contouring: current use, perceptions of and barriers to implementation. *Clin Oncol.* 2023;35(4):219–26. <https://doi.org/10.1016/j.clon.2023.01.014>.
104. Stroh L, et al. Implementation of artificial intelligence (AI) applications in radiology: hindering and facilitating factors. *Eur Radiol.* 2020;30(10):5525–32. <https://doi.org/10.1007/s00330-020-06946-y>.
105. Chua IS, et al. Artificial intelligence in oncology: path to implementation. *Cancer Med.* 2021;10(12):4138–49. <https://doi.org/10.1002/cam4.3935>.
106. Victor Mugabe K. Barriers and facilitators to the adoption of artificial intelligence in radiation oncology: a New Zealand study. *Tech Innov Patient Support Radiat Oncol.* 2021;18:16–21. <https://doi.org/10.1016/j.tipsro.2021.03.004>.
107. Vigdorovits A, et al. Breaking barriers: AI's influence on pathology and oncology in resource-scarce medical systems. *Cancers.* 2023;15(23):5692.

108. Blake H. AI in Healthcare: ethical considerations in the use of AI for diagnosis, treatment, and patient data management. 2024.
109. Gerke S, Minssen T, Cohen G. Chapter 12—Ethical and legal challenges of artificial intelligence-driven healthcare. In: Bohr A, Memarzadeh K, editors. *Artificial intelligence in healthcare*. Cambridge: Academic Press; 2020. p. 295–336.
110. Pesapane F, Summers P. Chapter 12—Ethics and regulations for AI in radiology. In: Ben-David S, editor. *Artificial Intelligence for Medicine*. Cambridge: Academic Press; 2024. p. 179–92.
111. Ahmad Z, et al. Artificial intelligence (AI) in medicine, current applications and future role with special emphasis on its potential and promise in pathology: present and future impact, obstacles including costs and acceptance among pathologists, practical and philosophical considerations. A comprehensive review. *Diagn Pathol*. 2021;16(1):24. <https://doi.org/10.1186/s13000-021-01085-4>.
112. Alaqra AC, Broden K, Bellström P, Alaqra AS. AI adoption in oncology: implementation process, barriers, and facilitators. *Elected Papers of the IRIS*. 2024;2024(15):5.
113. Khanna S, et al. AI governance framework for oncology: ethical, legal, and practical considerations. *Quart J Comput Technol Healthcare*. 2021;6(8):1–26.
114. Shreve JT, Khanani SA, Haddad TC. Artificial intelligence in oncology: current capabilities, future opportunities, and ethical considerations. *Am Soc Clin Oncol Educ Book*. 2022;42:842–51. https://doi.org/10.1200/EDBK_350652.
115. Pesapane F, et al. Chapter 52—Regulatory issues for artificial intelligence in radiology. In: Faintuch J, Faintuch S, editors, et al., *Precision medicine for investigators, practitioners and providers*. Cambridge: Academic Press; 2020. p. 533–43.
116. Subasi ID, Özçelik ŞB. Artificial intelligence in breast imaging: opportunities, challenges, and legal-ethical considerations. *Eurasian J Med*. 2023;55(1):114–9. <https://doi.org/10.5152/eurasianjmed.2023.23360>.
117. Odle T. The AI Era: the role of medical imaging and radiation therapy professionals. *Radiol Technol*. 2020;91(4):391–400.
118. Sasidharan Pillai A. Utilizing deep learning in medical image analysis for enhanced diagnostic accuracy and patient care: challenges, opportunities, and ethical implications. *J Deep Learn Genom Data Anal*. 2021;1(1):1–17.
119. Hantel A, et al. Perspectives of oncologists on the ethical implications of using artificial intelligence for cancer care. *JAMA Netw Open*. 2024;7(3):e244077–e244077. <https://doi.org/10.1001/jamanetworkopen.2024.4077>.
120. Wang P, et al. Ethics and safety in medical imaging and artificial intelligence. In: Liu S, editor, et al., *Artificial intelligence in medical imaging in China*. Singapore: Springer Nature Singapore; 2024. p. 431–46.
121. Cohen EB, Gordon IK. First, do no harm. Ethical and legal issues of artificial intelligence and machine learning in veterinary radiology and radiation oncology. *Vet Radiol Ultrasound*. 2022;63(S1):840–50. <https://doi.org/10.1111/vru.13171>.
122. Rowe SP, Pomper MG. Molecular imaging in oncology: current impact and future directions. *CA Cancer J Clin*. 2022;72(4):333–52. <https://doi.org/10.3322/caac.21713>.
123. Cobanaj M, et al. Chapter 8—Artificial intelligence in the oncology workflow: applications, limitations, and future perspectives. In: Ben-David S, editor, et al., *Artificial Intelligence for Medicine*. Cambridge: Academic Press; 2024. p. 91–111.
124. Katta MR, et al. Artificial intelligence in pancreatic cancer: diagnosis, limitations, and the future prospects—a narrative review. *J Cancer Res Clin Oncol*. 2023;149(9):6743–51. <https://doi.org/10.1007/s00432-023-04625-1>.
125. Alsharif F. Artificial intelligence in oncology: applications, challenges and future frontiers. *Int J Pharm Investig*. 2024;14:647–56. <https://doi.org/10.5530/ijpi.14.3.76>.
126. Corti C, et al. Artificial intelligence in cancer research and precision medicine: applications, limitations and priorities to drive transformation in the delivery of equitable and unbiased care. *Cancer Treat Rev*. 2023;112:102498. <https://doi.org/10.1016/j.ctrv.2022.102498>.
127. Alshuhri MS, et al. Artificial intelligence in cancer diagnosis: opportunities and challenges. *Pathol Res Pract*. 2024;253:154996. <https://doi.org/10.1016/j.prp.2023.154996>.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.